

Safety Is a Claim, Not a Cost: Calibrated Action Certificates for Offline RL under Moving Constraints

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Abstract

Offline safe reinforcement learning is usually evaluated by aggregate reward and cost under one or a few constraint budgets. This is an incomplete form of deployment evidence: a policy can satisfy an aggregate episode budget while relying on actions whose safety claims are unsupported under the requested deployment constraint. We study this deployability gap and propose action-conditional conformal certificates (ACCS), a post-hoc protocol that turns candidate policy actions into calibrated action-level safety claims with explicit provenance. ACCS builds a budget-uniform upper-cost certificate surface, executes or replaces actions whose claims are supported, and abstains when calibration evidence is insufficient. The intended contribution is not another constraint-adaptive policy optimizer, but a deployment evidence layer that can wrap offline safe RL methods, including strong policy-switching competitors.

Introduction

The current draft route is maintained in `../idea_blueprint.md`. This file is a format-valid AAAI-27 skeleton only; paper-facing claims must not be added before the Phase 0 literature gate and Phase 1 deployability-gap toy phenomenon are recorded.

Problem Setup

We consider an offline dataset collected under a behavior policy, a learned base policy, and deployment requests specified by cost budgets or safety constraints that may differ from those used during training or model selection.

Method

ACCS will be described as a budget-uniform claim-issuing protocol:

1. fit nonconformity scores for candidate actions on a held-out calibration split;
2. condition calibration by action groups, budget ranges, or risk clusters;

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3. execute, replace, or abstain according to the calibrated action-safety claim;
4. report certified utility, claim yield, claim miscoverage, unsupported safety success, budget adaptivity, and horizon-level audit metrics.

Experiments

The first experiment must establish a clean deployability-gap failure mode before broad benchmark runs. Direct competitors and reusable benchmark code are tracked in `../papers/literature_review.md`.

Conclusion

Placeholder.